

# LOKESH R M

AI / ML Engineer · Agentic AI, Computer Vision, Edge AI, Python

+91 9035446966 | [hi.lokeshrm@gmail.com](mailto:hi.lokeshrm@gmail.com) | Bengaluru, Karnataka, India

[linkedin.com/in/thisislokeshrm](https://linkedin.com/in/thisislokeshrm) | [github.com/thisislokeshrm](https://github.com/thisislokeshrm) | [thisislokeshrm.github.io](https://thisislokeshrm.github.io)

## SUMMARY

AI Engineer at Sirena Technologies building production agentic AI and robotics systems. Architected Agent OS, a 53 service multi agent platform running at 3 enterprise clients, and own the Policy Gateway enforcing 153 policies across it. Work spans LLMs, computer vision, voice pipelines and edge inference, with cloud services supporting 80+ concurrent robots.

## EXPERIENCE

**AI Engineer** - Sirena Technologies, Bengaluru, India

Jul 2026 - Present

- Architected **Agent OS**, a 53 service multi agent platform deployed to 3 enterprise clients, letting in house and third party agents interoperate to model business functions for small and medium sized businesses. Platform team of 12.
- Designed and own the **Policy Gateway**, enforcing 153 policies for centralised agent authorisation, control and monitoring. Led a team of 3.
- Led development of 5 in house agents across opportunity tracking, ROI, marketing and finance workflows, and 3 third party agent integrations.
- Led a 5 person team building **Sirena Cloud**, the voice, vision and LLM service layer supporting 80+ concurrent robots on an NVIDIA RTX Pro 6000.
- Built a streaming voice pipeline combining VAD, ASR, TTS and LLM reasoning at roughly 60 ms response latency.
- Fine tuned YOLO and SSD detectors for office and industrial environments, and optimise inference with ONNX for NVIDIA Jetson Orin Nano.
- Self host open weight LLMs, build RAG retrieval on pgvector, and deploy ROS 2 Nav2 and SLAM navigation for autonomous service robots.

**Robotics Engineer, Graduate Trainee** - Sirena Technologies, Bengaluru and Solan, India

May 2025 - Jun 2026

- Built MATLAB and Python locomotion simulations for biped and quadruped robots covering walk sequences and control algorithms, and validated every platform in simulation before hardware build.
- Developed features in the LS6 customised robot controller codebase and implemented ROS 2 motion control for robotic arms and wheeled robots.
- Led a 3 person team building the cloud system for an ESP32 P4 based home robotics product, including ESP-IDF firmware integration.
- Completed the graduate engineering rotation at the Solan facility and converted to a full time engineering role.

**Freelance Full Stack and AI Engineer** - Independent, Bengaluru, India

2024 - Present

- Delivered production web and AI applications for 5+ clients, covering LLM integration, agent workflows, APIs and backend infrastructure.

**Machine Learning Intern** - MURTI Labs, GITAM Bengaluru, India

May 2024 - Apr 2025

- Trained LSTM based object detection and action recognition models on drone video to 83% accuracy, and implemented path mapping for autonomous delivery and emergency response drones.
- Built the image processing and data collection pipeline feeding model training.

## SKILLS

**AI and Machine Learning:** LLMs, RAG, Multi Agent Systems, Agent Orchestration, PyTorch, TensorFlow, Scikit-learn, LSTM, pgvector, Open Weight Model Hosting

**Agent Frameworks:** LangGraph, LangChain, CrewAI, AutoGen, MCP

**Computer Vision and Speech:** YOLO, SSD, OpenCV, MediaPipe, ONNX, ASR, TTS, Voice Activity Detection

**Robotics and Edge AI:** ROS 2, Nav2, SLAM, NVIDIA Jetson Orin Nano, RTX Pro 6000, NVIDIA Isaac Sim, ESP32 P4, Edge Inference

**Infrastructure:** Docker, Kubernetes, FastAPI, Kafka, Redis, Google Cloud, PostgreSQL, MongoDB, Linux, Git

**Programming:** Python, C++, C, MATLAB, SQL, JavaScript

## PROJECTS

**Secure Federated Learning with Blockchain**

Sep 2024

Federated training system letting organisations train shared models on decentralised data, with blockchain backed coordination and data provenance so raw records never leave the owner.

Stack: Python, TensorFlow, Hyperledger Fabric, Solidity, Homomorphic Encryption, Differential Privacy

## EDUCATION

**B.Tech, Computer Science and Engineering** - GITAM (Deemed to be University), Bengaluru

Aug 2022 - Aug 2026

CGPA: 7.9 / 10

## ACHIEVEMENTS

- **INSPIRE Award MANAK** (Government of India, state level) · **Young Scientist Award** (Government of Karnataka, district level) · **Rajya Puraskar** (Governor's Award) · Finalist, Internal Smart India Hackathon, GITAM
- **Secretary, IEEE Computer Society GITAM** (Dec 2024 - Aug 2025) · Student Placement Coordinator, GITAM Career Guidance Cell